SATVIK GANDA

Computer Engineering Undergraduate • AI/ML • Software Development • Computer Vision

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PROFESSIONAL SUMMARY

Computer Engineering undergraduate with hands-on experience across software development, machine learning, deep learning, computer vision, and generative AI. Proficient across the full development lifecycle, from problem definition and data processing to model implementation, backend architecture, UI development, and deployment. Adept at learning new frameworks quickly and delivering technically ambitious, production-oriented systems.

EDUCATION

B.E. in Computer Engineering *Expected 2028*

Thapar Institute of Engineering & Technology, Patiala

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, DBMS, Computer Architecture & Organization, Discrete Mathematics, Probability & Statistics, Numerical Analysis, Artificial Intelligence & Machine Learning, Image Processing, Software Engineering.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, Dart, Kotlin, SQL

Development: React, React Three Fiber, Flutter, PHP, Streamlit

Databases: MySQL, Oracle / PL-SQL

Tools & Platforms: Git, GitHub, VS Code, Android Studio, Jupyter Notebook, MATLAB, R, Unity, Blender, Cisco Packet Tracer

AI / Machine Learning: Machine Learning, Deep Learning, PyTorch, NumPy, Data Preprocessing, Model Training & Evaluation, Sequence Modeling, Knowledge Tracing, Local LLMs, RAG, Embeddings, Vector Databases, AI Orchestration

Computer Vision & Data: OpenCV, Image Processing, Grayscale/Binary Conversion, Thresholding, Nearest-Neighbor & Bilinear Interpolation, Image Blending, Padding, Histogram Equalization & Matching, Numerical/Data Analysis

FEATURED PROJECTS

Sovara AI — AI / RAG Orchestration System | *Local LLMs, RAG, Qdrant, Embeddings*

- Designed a modular AI orchestration system integrating local LLMs, retrieval-augmented generation, embeddings, and Qdrant for grounded, context-aware responses.
- Implemented planning, tool execution, and multi-step workflow logic to support structured and reliable AI task execution.

Self-Attentive Knowledge Tracing | *PyTorch, Sequence Modeling*

- Implemented a self-attentive knowledge tracing model for student learning analytics using PyTorch and real-world educational interaction datasets.
- Built the data preprocessing and sequential input pipeline, followed by model training and performance evaluation.

Scene Aided Navigation System (SAN) | *PyTorch, OpenCV, Python, Satellite Imagery, Kalman Filter*

- Developed a visual geo-localization pipeline for GPS-denied aerial navigation — matching live nadir camera frames against pre-stored satellite map tiles using learned CNN descriptors and contrastive training.
- Built a simulation framework using real satellite imagery (Sentinel-2) to model airborne vehicle flight paths and evaluate localization accuracy across terrain types including urban, rural, and forested scenes.
- Implemented an Extended Kalman Filter (EKF) to fuse SAN position estimates with IMU data, achieving robust position tracking under sensor noise and occlusion conditions.
- Research conducted in alignment with DRDO DIA-CoEs problem statement (ADE Lab) — deliverables include source code, sensor configuration analysis, and test results for cruise missile / UAV applications in GPS-denied environments.

CampusVerse — Interactive Campus Experience Platform | *Unity, VR, UI/UX*

- Designed and developed the user interface for CampusVerse, focusing on a polished, intuitive experience for exploring campus-related information and interactive features.

- Developed a VR Lab Simulator in Unity, creating an interactive virtual laboratory environment to simulate academic lab experiences and enable immersive exploration.

Banknote Authentication ML System | *Python, Streamlit*

- Built an end-to-end machine learning application for banknote authentication, covering data preprocessing, model training, classification, and inference.
- Developed and deployed an interactive Streamlit dashboard enabling users to input features and obtain real-time authentication predictions.

Student Registration Application | *Flutter, PHP, MySQL*

- Developed a full-stack student registration application with a Flutter frontend and PHP backend for handling user-facing registration workflows.
- Integrated the frontend with server-side data processing and database operations to manage student information.